International Journal of Energy and Water Resources (2026) 10:37
https://doi.org/10.1007/s42108-025-00450-z

ORIGINAL ARTICLE

Numerical analysis of flow characteristics through stilling basin at pump stations outlets

Elsayed M. Elshahat · Mohab Elhousseiny[1] · Maha R. Fahmy[1] · Hany F. Abd-Elhamid[1,2] · Martina Zeleňáková[3] · Mohamed Awad[1]

Received: 6 September 2025 / Revised: 22 October 2025 / Accepted: 2 November 2025
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Abstract
Stilling basins are critical techniques in hydraulic engineering that are designed to dissipate the kinetic energy of water flowing from pipe outlets. This study aims to assess the performance of stilling basin using ANSYS Fluent focusing on the effects of varying the height and position of the impact wall from the pipe outlet on energy dissipation. A parametric study was conducted to investigate significant parameters including the relative energy dissipation, average longitudinal velocity, turbulent dissipation rate, and turbulent kinetic energy. The findings indicate that the in-practice basin exhibits a 0.37% higher relative energy dissipation compared to the United States Bureau of Reclamation basin. Reducing the impact wall distance from pipe outlet is accompanied by an increase in energy dissipation, with the highest dissipation observed at a wall of distance 3d (d is the pipe outlet diameter). At this distance, dissipation is 1.5% greater than that recorded at 2d. Energy dissipation is also influenced by the height of the impact wall. Higher walls (3.167d and 3.75d) achieve more energy dissipation. Also, the in-practice basin also showed a smoother velocity profile, with the velocity decreasing significantly after encountering the impact wall. The maximum values of turbulent dissipation rate and turbulent kinetic energy were located near the impact wall, especially with greater heights. These results offer valuable insights for designing the stilling basin to better control flow. The outcomes can improve the design of pump station outlets and are also globally applicable.

Keywords Baffle wall · Energy dissipation · Jet flow · Pipe outlet · Stilling basin · ANSYS fluent

✉ Hany F. Abd-Elhamid
hany.abdelhamid@tuke.sk; hany_farhat2003@yahoo.com

Elsayed M. Elshahat
eng_sayedzaki@yahoo.com

Mohab Elhousseiny
mohabhazem@eng.zu.edu.eg

Maha R. Fahmy
maharashad@yahoo.com

Martina Zeleňáková
martina.zelenakova@tuke.sk

Mohamed Awad
Eng_awad2004@yahoo.com

[1] Department of Water and Water Structures Engineering, Zagazig University, Zagazig 44519, Egypt

[2] Department of Environmental Engineering, Faculty of Civil Engineering, Technical University of Kosice, 040 01 Košice, Slovakia

[3] Institute of Sustainable and Circular Construction, Faculty of Civil Engineering, Technical University of Kosice, 040 01 Košice, Slovakia

Abbreviations

USBR	United States Bureau of Reclamation
TKE	Turbulent Kinetic Energy
CFWJ	Counter-flowing wall jet
CFFJ	Counter-flowing free jet
PWD	Public Works Department
WES	Water and Environmental Services
CFD	Computational fluid dynamics

Introduction

Water usually flows with high kinetic energy and small depths at the downstream of hydraulic structures, such as weirs and pumps stations, endangering the stability of these facilities and causes scour formation on the channel bed. Thus, energy dissipation of excess kinetic energy is at the forefront of open channel topics to restore the normal flow conditions at the downstream reach (Atkins, 1994, Paul Peter Urone, 2022, Kheirehdast & Ghodsian, 2024). Generally, there are three techniques that can be utilized for energy

Published online: 22 December 2025

dissipation: (a) stilling basins; (b) counter-flow or reverse flow; and (c) slopping surfaces (Mohammed et al., 2020). They all include appurtenances that furtherly contribute to drag, turbulence, friction and energy dissipation (Mei et al., 2025; Vischer, 2018). For hydraulic structures, stilling basins are the most used technique. Between the 1940s and the 1960s, several standard stilling basins were developed and tested by models and prototypes over a considerable range of operating flow conditions. Since then, different types of stilling basins have been introduced (Riad, 2008, Hager, 2013, Abbas et al., 2018, Hager, 2018, Rice & Kadavy, 1993, Kamaraj & Rangarajan, 2022, Jain & Jacob, 1970, Varshney and International, 1965, Corry, 1975). Appurtenances such as sills, steps, baffles, piers, and chute blocks are used at these basins to control the hydraulic jump, increase efficiency and shorten the basin's length (Mardani et al., 2015). Therefore, the proper type is selected upon various factors including the water structure itself, the required dissipation rate and the flow conditions, in addition to, practical imperatives and industry exigencies. Similarly, water jets at pump stations outlets discharge with extremely high velocities. Therefore, almost identical techniques are used to ensure the appropriate dissipation of these jets. In the counterflow method, different configurations can be applied, sharing the same concept of the collision process depending on splitting the flow into two opposing streams. Counter-flowing wall jets (CFWJ) and counter-flowing free jets (CFFJ) are two types of counter-flowing jets that have recently gained increased use (Sharma et al., 2019). The theoretical equations for estimating conjugate depths, energy loss and forced hydraulic jump characteristics were predicted in the analytical study of (Stevović, 2016). Further various counterflow configurations were investigated through implementing and testing different physical models in labs (Alghwail et al., 2018; Hager, 1994; Vollmer & Khader, 1971; Wang et al., 2022). An additional widely employed method for energy dissipation involves the use of inclined or stepped basins, whose efficiencies have been extensively investigated across various basin configurations (Arjenaki & Sanayei, 2020; Chanson, 2000; Pagliara et al., 2015; Salmasi & Abraham, 2022a, 2022b; Shahheydari et al., 2015).

Literature review

Among the techniques discussed, stilling basins for pipe outlets have remained the most widely adopted and preferred method to date. They originated in the nineteenth century with a variety of designs, including PWD basin, WES basin, hump type stilling basin, manifold stilling basin, contra costa stilling basin, Hook basin, Utah state university stilling basin, counterflow stilling basin, Garde's stilling basin and USBR VI basin (Chanson, 2002, Gehlot & Tiwari, 2014, rasheed, 1963, Hager, 2010, Goel, 2008, Hager, 2013, Fiala & Albertson, 1963, Hager & Dams, 1996, Flammer et al., 1970, Keim, 1962). Each of these types has a unique dissipator which differentiates it from the others, as summarized in Table 1. This table offers a concise review of prior research on stilling basins for pipe outlets. The USBR VI stilling basin is one of the widely used basins, discovered by the United States Bureau of Reclamation, for pipe outlets as shown in Fig. 1. Most networks for surface water collection, irrigation, and drainage use this basin for inflow Froude number ranges between 1 and 10. It relies heavily on the concept of impaction, where impact forces and shear stresses are exerted on the system when the flowing jet in the channel encounters an obstacle, regardless of its shape or size (Tiwari et al., 2022). The lateral distribution of water jet along the entire width of the basin ensures a proper dissipation and leads to an efficient, economical energy dissipator (Verma & Goel, 2003). Recently, two notches are added to the baffle bottom to prevent the basin from filling with sediments. All previous experimental studies addressed the functionality of the stilling basin for circular pipe outlets, particularly the Garde and USBR VI basins, and focused on improving their performance (Arun Goel & Verma, 1999; Goel, 2008; Goel & Verma, 2006; Pillai et al., 1989; Verma & Goel, 2003). However, Tiwari with others focused on investigating non-circular pipe outlets. Their findings were published in several articles (Tiwari et al., 2011, Tiwari and Seema, 2013b, Tiwari and Seema, 2013a, Tiwari, 2013, Tiwari & Goel, 2014, Tiwari & Goel, 2016, Tiwari & Singh, 2017). Due to the complexity of the dissipation process occurring within the basins, physical models lack a comprehensive analysis of the flow field. To this end, there is now interest in research into using CFD software (Wei-lin et al., 1996, Lu et al., 2008, Kannan et al., 2017b, Al-Husseini et al., 2018, Babaali et al., 2019, Mohammed et al., 2020, Sharma et al., 2021, Behnamtalab et al., 2022, Behnamtalab et al., 2023, Farouk et al., 2022).

Lately, Pump stations have become one of the most significant hydraulic structures in Egypt, owing to their critical role in the country's major agricultural and urban development projects. Such examples include the huge land reclamation projects in North Sinai, New Delta, Mustaqbal Misr, Tushki and West Noubaria Rural Development. Despite all the previously mentioned basins, the stilling basins downstream of pump stations in these projects are implemented on a practical basis and differ significantly from the standard basin designs discussed earlier. Most of these basins are attributed by a baffle wall over which the jet flows, either it is placed perpendicular to the flow direction or inclined. Although the wide use of them at Egypt's pump stations outlets, there is no previous reports on them in the main literature according to author knowledge. This study clearly investigates the numerical analysis of an in-practice stilling

Table 1 Summary of literature review on stilling basins for pipe outlet and their optimization

Author	Method	Stilling basin type	Input parameters	Output parameters
(Pillai et al., 1989)	Experimental	Circular pipe outlet	Various splitter and end sill shapes	Using rounded end sill and wedge-shaped splitter with a vertex angle of 150° makes the basin more efficient
(Goel & Verma, 2006)	Experimental	Circular pipe outlet	Wedge-shaped splitter block, a modified grid, an intermediate sill and a rounded step	Compared to Garde's model of 12d length for Froude no. 5.50, the basin length can be reduced by 50% with better performance
(Tiwari et al., 2011)	Experimental	Non-circular pipe outlet	Rectangular, square and sloping end sill for Froude no. 1.85, 2.85 and 3.85	For all Fr, the sloping end sill (1 V:1 H) dissipates more energy
(Tiwari, 2013)	Experimental	Non-circular pipe outlet	Optimizing USBR VI basin by varying geometric properties of impact wall for Fr 1.85, 2.85 and 3.85	the impact wall of dimensions 1.5d × 3d is more efficient and its location affects the scour index
(Tiwari & Seema, 2013a)	Experimental	Non-circular pipe outlet	Rectangular, square and different sloping end sill (1 V:1 H), (1 V:0.5 H), (1 V:1.5 H), (1 V:2 H) for Froude no. 1.85, 2.85 and 3.85	The maximum dissipation is obtained using the sloping end sill (1 V:1 H)
(Tiwari & Seema, 2013b)	Experimental	Non-circular pipe outlet	Different shapes and geometric properties of intermediate sill for Fr 1.85 and 3.85	the square intermediate sill improves the efficiency of the basin
(Tiwari & Goel, 2014)	Experimental	Non-circular pipe outlet	Rectangular, square, triangular and trapezoidal end sill for Froude no. 1.85, 2.85 and 3.85	The triangular end sill (1 V:1 H) is more efficient
(Tiwari et al., 2014)	Experimental	Non-circular pipe outlet	Optimizing USBR VI basin by involving different shapes of intermediate sill	Inclusion of square intermediate sill ensures higher performance
(Tiwari et al., 2015)	Experimental	Non-circular pipe outlet	Inverted T-Shape splitter blocks of different dimensions for Fr 2.85 and 3.85	The performance improved by 5–6 times
(Sobhan Aleyasin et al., 2015)	Experimental	Circular pipe outlet	Adding flow splitters and cellular baffles to USBR VI basin	The splitter with baffle improves the efficiency of the basin
(Tiwari & Goel, 2016)	Experimental	Non-circular pipe outlet	Different locations and gap distance of impact wall for Fr 1.85, 2.85 and 3.85	An enhancement in basin efficiency and reduction in its length
(Tiwari & Singh, 2017)	Experimental	Non-circular pipe outlet	Different configurations of impact wall with square intermediate sill	The basin efficiency improved to more than 2 or 3 times the USBR VI basin
(Mohammed et al., 2020)	Numerical	Circular pipe outlet	Various configurations of dissipators and different inlet pipe height	The dissipator (L-section) dissipates more energy and the rectangular shape dissipates less
(Sharma et al., 2021)	Numerical	Non-circular pipe outlet	Impact wall size of 1.5d × 3d with variable location from pipe outlet and gap distance from bottom floor	Impact wall at distance 4d from pipe outlet and a gap of 1d has a higher dissipation rate
(Behnamtalab et al., 2022)	Numerical	Circular pipe outlet	$\frac{W}{d}$ from 9.23 to 3.50	A decrease in average outlet velocity and increase in EG with increasing $\frac{W}{d}$ ratio
(Tiwari et al., 2022)	Experimental	Non-circular pipe outlet	Addition of a triangular splitter block with an end sill and impact wall for Fr 1.85, 2.85 and 3.85	A reduction of basin's length by 29%
(Behnamtalab et al., 2023)	Numerical	Circular pipe outlet	$\frac{W}{d}$ from 9.23 to 3.50	The turbulent energy dissipation rate in the basin improves to 4.5 by increasing $\frac{W}{d}$ from 3.50 to 9.23

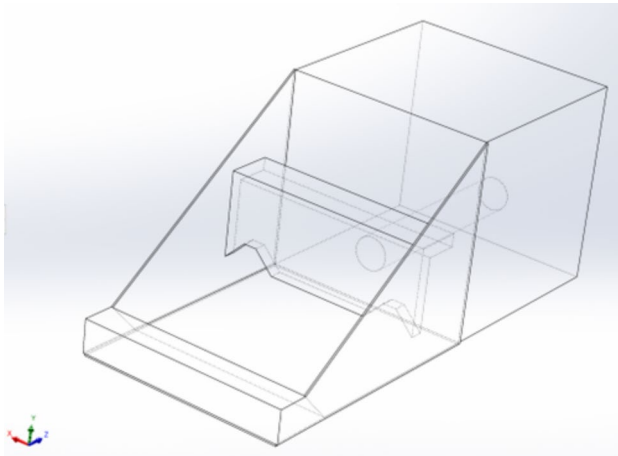


Fig. 1 Stilling basin for pipe outlets

by utilizing a parametric analysis centered upon the impact wall height and position. Comparatively evaluating these schemes against the traditional USBR VI basin, the investigation successfully bridges the gap between theory and practice. The use of advanced numerical tools to analyze crucial parameters such as relative energy dissipation, average longitudinal velocity, turbulent dissipation rate, and turbulent kinetic energy distinguishes this study from earlier research, offering better insights into the design optimization of hydraulic basins. This study was conducted in Egypt, commencing in 2023, with the aim of investigating the flow regimes through the outlets of the pump stations by means of stilling basins.

Materials and methods

The approach adopted in the current work is presented in the order mentioned (Fig. 2). The present study aims to develop a numerical model of a recently implemented in-practice basin featuring an impact wall positioned at a designated distance from the pipe outlet and set at a defined elevation. This model has been implemented using a Computational Fluid Dynamics (CFD) approach based on the ANSYS FLUENT 2020R2 software package. This approach compares the energy dissipation and efficiency of this design with the traditional USBR VI basin, which is extensively used in hydraulic engineering. This software is renowned for its capability in the simulation of turbulent flow, and it offers state-of-the-art turbulence models essential for making credible predictions of flow behavior for such hydraulic systems. The methodology is organized in a sequential flow, which includes the following stages: pre-processing, simulation/solution, and post-processing. The steps are designed to ensure that the study captures the key physical phenomena involved in the flow through the basin, evaluates energy dissipation rates, and allows a meaningful comparison between the traditional and the in-practice basin ['Experimental Validation of CFD Modeling of Multiphase Flow through Open Channel Confluence') (Jeong & Seong, 2014) (ANSYS, 2009).

Governing equations

The following equation is defined as the conservation of mass, or continuity equation (ANSYS, 2009):

$$\frac{\partial \rho}{\partial t} + \nabla\left(\rho \vec{v}\right) = S_m \tag{1}$$

It represents the general form of the mass conservation equation and is applicable for both incompressible and compressible flow. The term S_m refers to added mass to the current phase from the dispersed other phase and ρ represents the fluid mass and $\vec{v}$ is the velocity. While the flow in the current study is incompressible and there is no source of mass existed, then the reduced continuity equation takes the form (ANSYS, 2009):

$$\frac{\partial}{\partial x}\left(uA_x\right) + \frac{\partial}{\partial y}\left(vA_y\right) + \frac{\partial}{\partial z}\left(wA_z\right) = 0 \tag{2}$$

u, v and w are instantaneous velocities in X, Y and Z directions, A_x, A_y and A_z are cross-sectional areas in X, Y and Z.

For the momentum conservation (Averaged Navier Stokes) equations, it is stated as (ANSYS, 2009):

$$\frac{\partial}{\partial t}\left(\rho\vec{v}\right) + \nabla.\left(\rho\vec{v}\vec{v}\right) = -\nabla p + \nabla.\left(\bar{\bar{\tau}}\right) + \rho\vec{g} + \vec{F} \tag{3}$$

$\rho\vec{g}$ and $\vec{F}$ are the gravitational body force and external body forces, respectively. p is the static pressure and $\bar{\bar{\tau}}$ is the Reynolds stress tensor, which is measured from Eq. 4 (ANSYS, 2009):

$$\bar{\bar{\tau}} = \mu\left[\left(\nabla\vec{v} + \vec{v}^T\right) - \frac{2}{3}\nabla.\vec{v}I\right] \tag{4}$$

μ is the molecular viscosity and I is the unit tensor.

The turbulent kinetic energy (TKE) and turbulent dissipation rate are key parameters that strongly indicate the magnitude of energy dissipation. TKE is expressed as the mean kinetic energy per unit mass and represents the energy associated with the turbulent motion of the fluid. In turbulent flow, eddies are developed, and water velocity fluctuations occur throughout the fluid domain. TKE can be calculated using the following equation (Rehman & Hong, 2022):

$$TKE = \frac{1}{2}\left(\overline{(u')^2} + \overline{(v')^2} + \overline{(\omega')^2}\right) \tag{5}$$

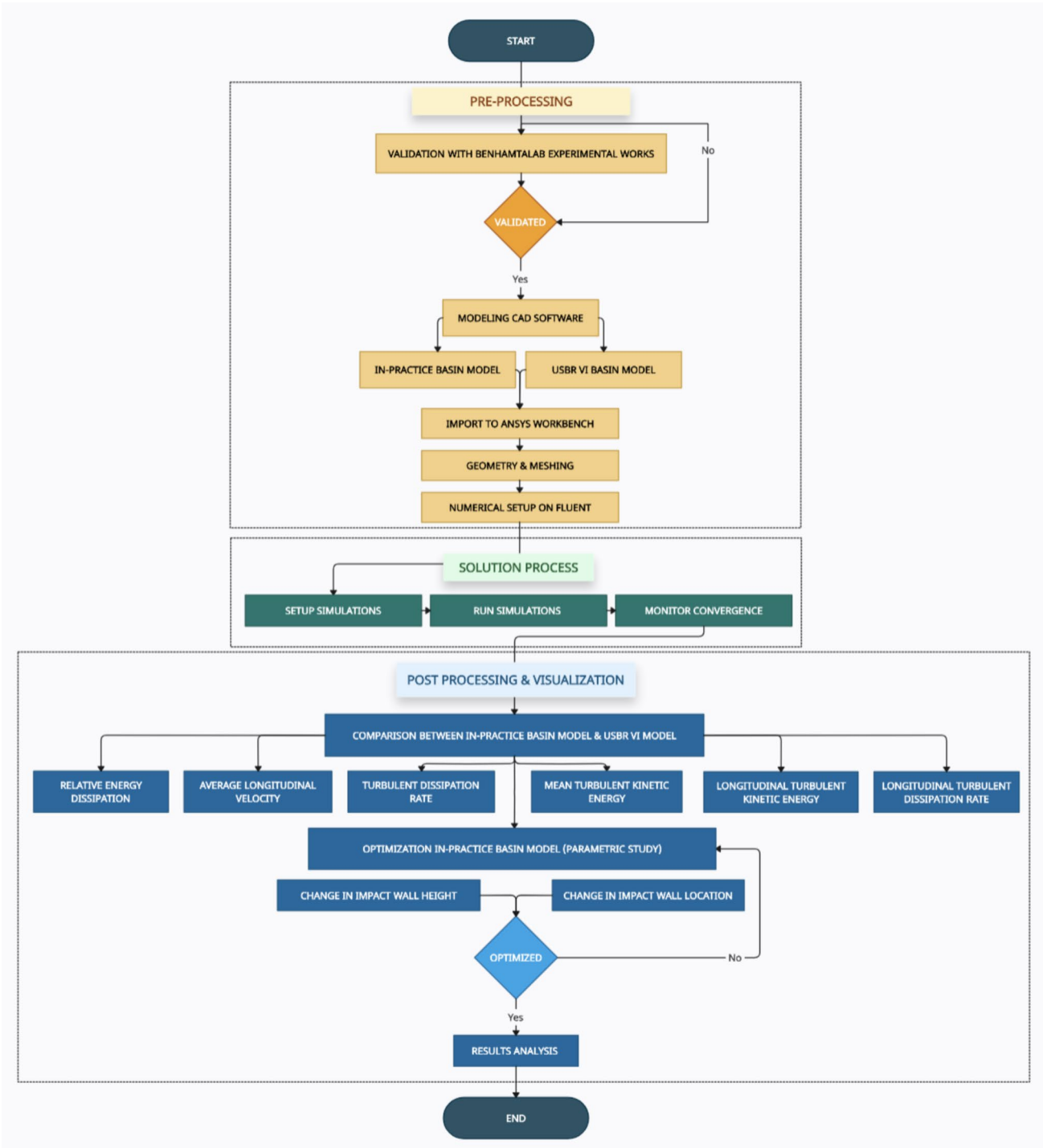


Fig. 2 Flow chart of the methodology used in the current study

$u\prime$ is the turbulent velocity component in X-direction, $\upsilon\prime$ is the turbulent velocity component in Y-direction and $\omega\prime$ is the turbulent velocity component in Z-direction.

The turbulent velocity is the difference between instantaneous and average velocity as shown Eq. 7.

$$u\prime = u - \overline{u} \tag{6}$$

u is the instantaneous velocity and $\overline{u}$ is average velocity.

Model setup and preprocessing

First, SOLIDWORKS Premium 2021 SP3.1 is used to create the geometry of USBR VI conventional basin. In addition, a comparative model of in-practice basin, including the impact wall at a specified height and distance from the pipe outlet. The inlet pipe diameter "d" was chosen to be 0.093 m (Sharma et al., 2021). The geometric features derived from actual dimensions and experimental data form the basis of the computational domain for both basins. The basin models are rectangular boxes with a length of 7d (65.1 cm), a width of 6.3d (58.8 cm) and a height of 5.2d (55 cm), where 'd' is the pipe diameter as shown in Fig. 3. In order to simulate the turbulence effects near the walls and predict flow behavior under high Reynolds number accurately, the simulations make use of RNG K-ε model with enhanced wall treatment as per the recommendations of earlier research (Kannan et al., 2017a, Choi et al., 1994, Papageorgakis & Assanis, 1999). Because of its accuracy and resilience to a variety of flow configurations, particularly in complex fluid dynamics scenarios like those found in stilling basins, this turbulence model was selected.

The computational domain was extended downstream of the basin until a fully developed uniform flow was achieved. To ensure full flow development at the pipe outlet or basin entrance, the length of the inlet pipe upstream was also accounted for in the model. Consequently, the inlet pipe length to the basin in models equals 3 m based on the relationship of the boundary layers in turbulent flow (Beichley, 1971). While the basin width are chosen (W = 6.3d) (Garde et al., 1990) (Brevard, 1971). Froude number of the water flowing from pump stations outlet was calculated based on Eq. (7) presented below (Beichley, 1971).

$$D = \sqrt{A} = \sqrt{\frac{Q}{\overline{u}}} \tag{7}$$

D is equivalent depth of flow at the basin entrance; A is cross-sectional area of the flow at the basin entrance; Q is discharge or volumetric flow rate and $\overline{u}$ refers to mean velocity at the basin entrance.

$$F_r = \frac{\overline{u}}{\sqrt{g.D}} \tag{8}$$

F_r denotes Froude number which is dimensionless parameter indicating flow regime and g is a gravitational acceleration (typically 9.81 $m/\sec^2$)

$$\frac{W}{D} = 2.86 \times F_r^{0.575} \tag{9}$$

W is stilling basin width. These equations and variable definitions are consistent with classical hydraulic research and design guidelines for energy dissipators and stilling basins.

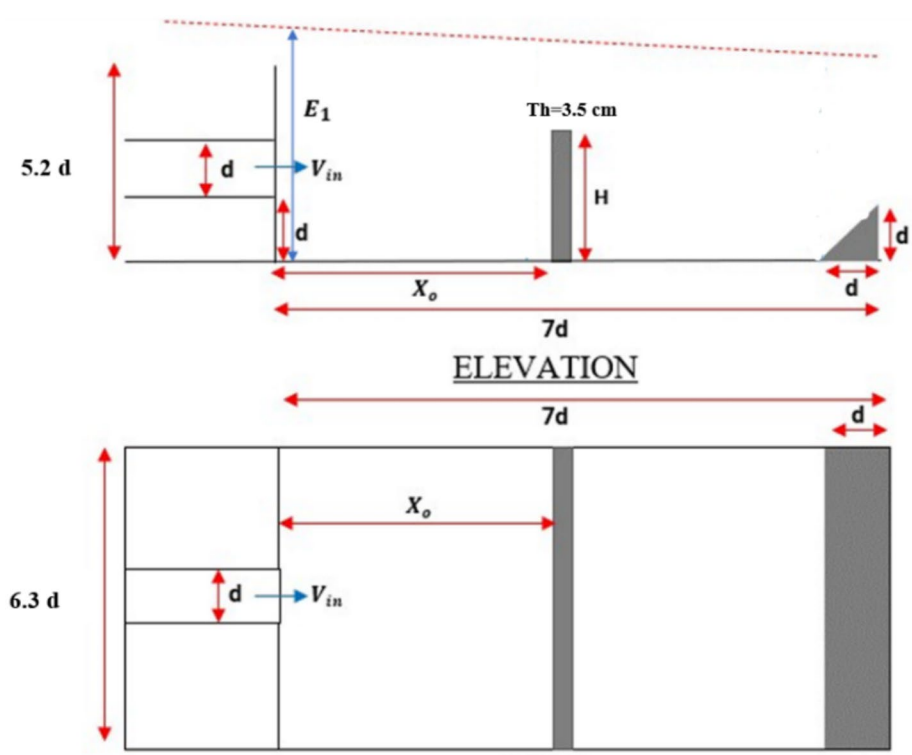


Fig. 3 Schematic representation of the geometric properties of the in-practice basin model

Meshing and boundary conditions

A structured mesh comprising approximately 1,935,006 elements and 363,743 nodes was employed, with refinement concentrated near the impact wall and in regions of high velocity and turbulence gradients as shown in Fig. 4. Mesh sensitivity analysis is performed by the GCI method to determine the optimal mesh size, which would ensure the solution is independent of the mesh size. Boundary conditions are clearly defined to include realistic operating conditions. An inlet velocity is defined at the inlet boundary to simulate flow from the pipe into the basin. The outlet boundary is simulated with pressure-outlet condition. The impact wall is defined as a no-slip wall.

Simulation scenarios

The study is divided into different simulation scenarios, starting with evaluating the performance of the in-practice basin compared to the USBR VI basin using ANSYS FLUENT 2016R2. A comprehensive parametric study is then conducted to investigate the effects of varying the height and distance of the impact wall in this basin as shown in Table 2. The key parameters presented in this table include the impact wall distance, baffle wall height, and the corresponding Froude number, which provide details on the flow regime under the different conditions. The location of the impact wall with respect to the pipe outlet is critical because it determines where the impact wall must be positioned in terms of the pipe outlet and directly affects the flow characteristics and energy loss of the system. In all scenarios, both the distance of the impact wall from the pipe outlet and the height of the impact wall were varied to observe their effects on the system's behavior. The results of such simulations are compared across a range of configurations to determine the optimal design for maximum energy dissipation. The following parameters are investigated to assess the basin's hydraulic performance (relative energy dissipation, average longitudinal velocity, turbulent dissipation rate and turbulent kinetic energy). Table 2 summarizes the different scenarios considered in this study, with variations in baffle wall distance and height.

Table 2 Different scenarios conducted in the current study

Scenario	Distance of baffle wall from pipe outlet (m)	Height of baffle wall (m)	Froude no dimensionless
Z-1	2d = 0.186	2d = 0.186	3.85
Z-2	2d = 0.186	2.29d = 0.213125	3.85
Z-3	2d = 0.186	2.58d = 0.24025	3.85
Z-4	2d = 0.186	2.875d = 0.267375	3.85
Z-5	2d = 0.186	3.167d = 0.2945	3.85
Z-6	2d = 0.186	3.458d = 0.321625	3.85
Z-7	2d = 0.186	3.75d = 0.34875	3.85
Z-8	2d = 0.186	4.042d = 0.375875	3.85
Z-9	2d = 0.186	4.33d = 0.403	3.85
Z-10	2d = 0.186	3.2d = 0.2976	3.85
Z-11	2.25d = 0.20925	3.2d = 0.2976	3.85
Z-12	2.5d = 0.2325	3.2d = 0.2976	3.85
Z-13	2.75d = 0.25575	3.2d = 0.2976	3.85
Z-14	3d = 0.279	3.2d = 0.2976	3.85
Z-15	3.25d = 0.30225	3.2d = 0.2976	3.85

Computational methods and convergence criteria

ANSYS FLUENT employs a pressure-based solver in this study. This solver is appropriate especially for incompressible fluid, which is often the case in hydraulic and civil engineering applications. It is also employed to determine the pressure and velocity fields, thereby providing an accurate

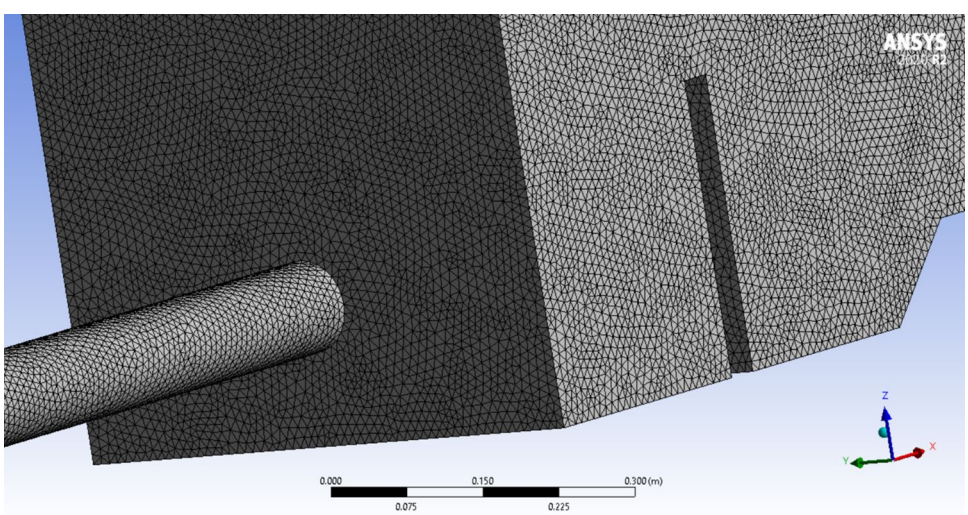


Fig. 4 Three-dimensional view of the computational domain showing the structured mesh distribution

simulation of fluid flow. The SIMPLE algorithm is applied in pressure velocity coupling, and second-order discretization schemes are used to ensure the accuracy of the solution. The transient formulation is applied in simulations to account for time-dependent behaviors of the flow. This can be applied in dynamic flow, which varies with velocity, pressure, and turbulence variables. Convergence is checked based on residuals of important flow variables (velocity, pressure, turbulence values) and physical values such as energy dissipation. A solution is said to be converged when the residuals go below a specified threshold and when observed flow parameters converge within a reasonable range (ANSYS, 2016).

Validation and comparison with experimental data

The numerical model's accuracy is validated by comparing it with an experimental study (Behnamtalab et al., 2019), which studies the USBR VI basin in hydraulic lab. Water depths were measured at various locations within the basin. The experimental data were collected from multiple points along the channel width, including the end sill, which is a key feature of the stilling basin. These measured water depths are compared with the obtained values from the numerical simulations. Figure 5 depicts how both the computational results and the experimental results are in good agreement, demonstrating that the model is capable of predicting the flow characteristics and energy dissipation in the basin. Also, the pressure head at a certain point of the impact wall was experimentally and numerically measured. While the experimental pressure head was 67 cm, the numerical model estimated it at 72 cm. The noted difference of 5 cm may be attributed to several factors, including the mesh's resolution and the uncertainties inherent in the measurement of physical experiments. The comparison nevertheless confirms for us that the numerical model accurately reflects the behavior recorded during the physical experiments, with the results falling within a tolerable accuracy range for the application under consideration. Quantitative goodness-of-fit metrics were further calculated for the purposes of furthering the validation process corresponding to the water depth profile comparison. The coefficient of determination (R^2) achieved a value of approximately 0.96, indicating a strong correlation and predictive capability of the numerical model for the data points measured. These key experimental data points, sourced from (Behnamtalab et al., 2019), provide strong evidence that the numerical model delivers reliable and accurate predictions of flow behavior within the basin. Furthermore, different turbulence models were tested in our case study. Enhanced wall function RNG $K-\varepsilon$ showed the best results with experimental data.

Results and discussion

Simulations conducted on a practical stilling basin and the standard USBR VI basin have demonstrated a significant capacity for energy dissipation. 15 different impact wall configurations, varying in height and distance from the pipe outlet, were tested. These configurations directly relate to the results shown in Figs. 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 and 16.

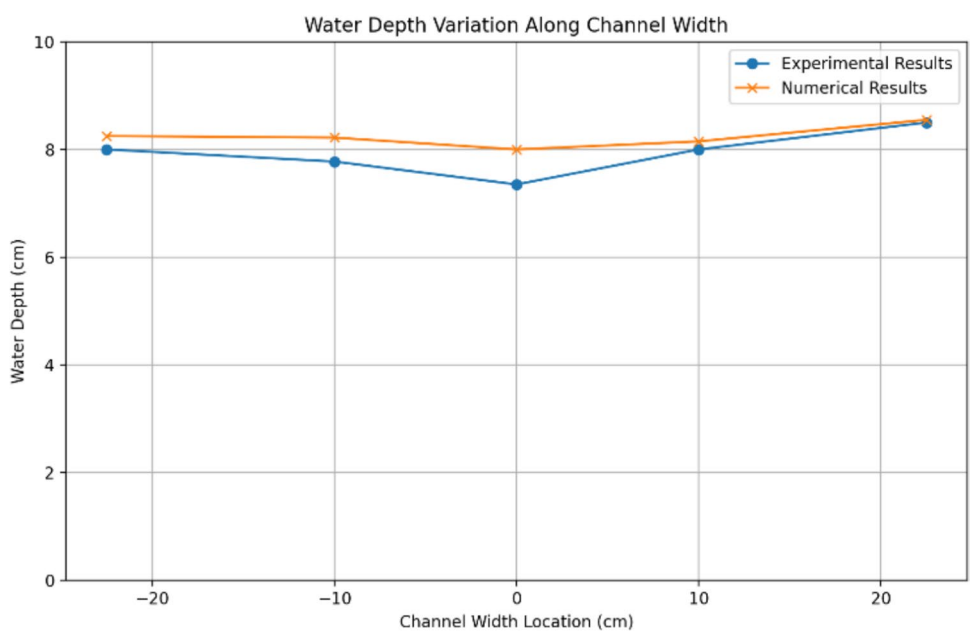


Fig. 5 Water depths at end sill across the channel width

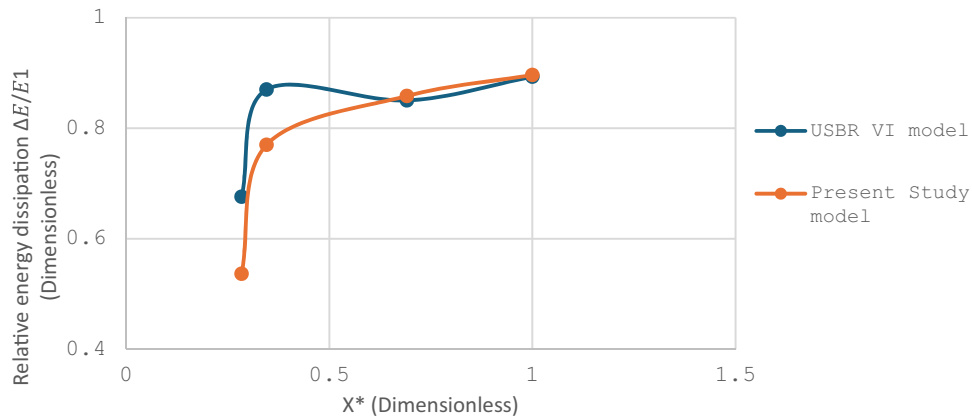


Fig. 6 Relative energy dissipation for USBR VI basin and developed basin

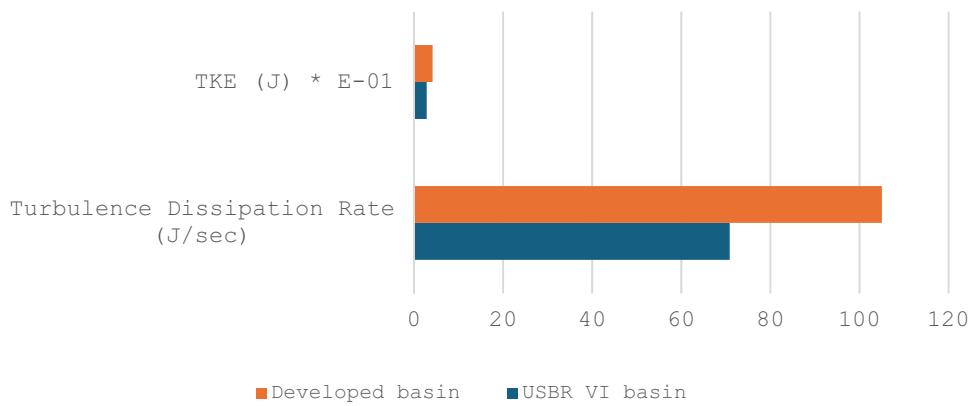


Fig. 7 Comparison between USBR VI model and the present study model according to turbulence quantities

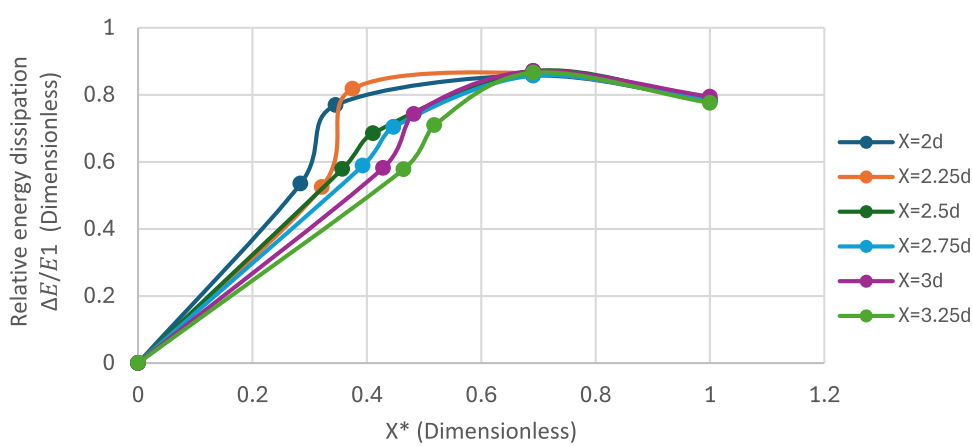


Fig. 8 Relative energy dissipation for different impact wall distances

Energy dissipation analysis

An initial focus is placed on the dissipation of energy. When the water jet exits from the pipe outlet, its velocity decreases, and the flow depth increases as a result of the impact upon the baffle wall. Following impaction, the water flows over the baffle with reduced kinetic energy, continuing until it reaches the end sill, which has a slope of 1:1. This sill facilitates a gradual energy dissipation.

Figure 6 demonstrates the relative energy dissipation for both the USBR VI basin and the in-practice basin. On the y-axis, there is a relative energy dissipation (dimensionless parameter) $\frac{\Delta E}{E_1}$, which represents the percentage of energy dissipated compared to the initial energy of the flow. The x-axis represents the relative position of the flow, specifically the distance along the basin relative to the whole basin length. The relative position is denoted

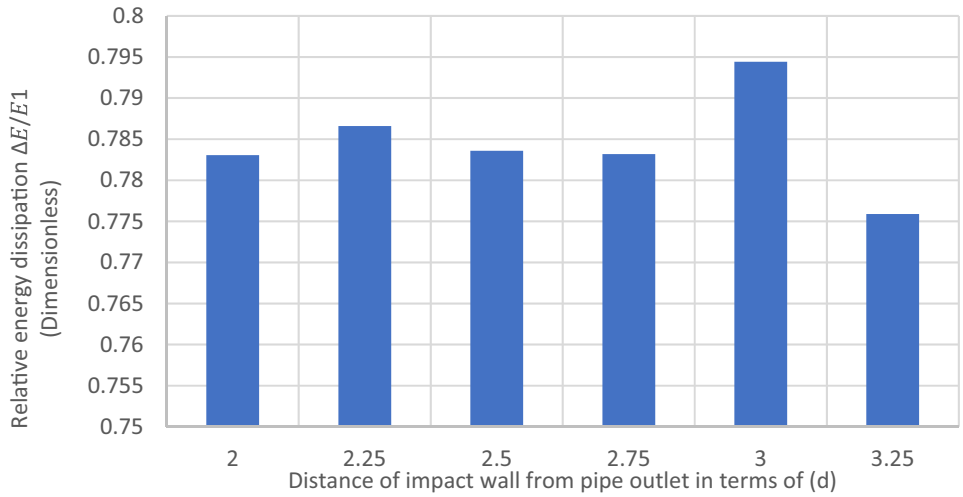


Fig. 9 Relative energy dissipation for different distances of impact wall at end sill of basin for Froude number 3.85

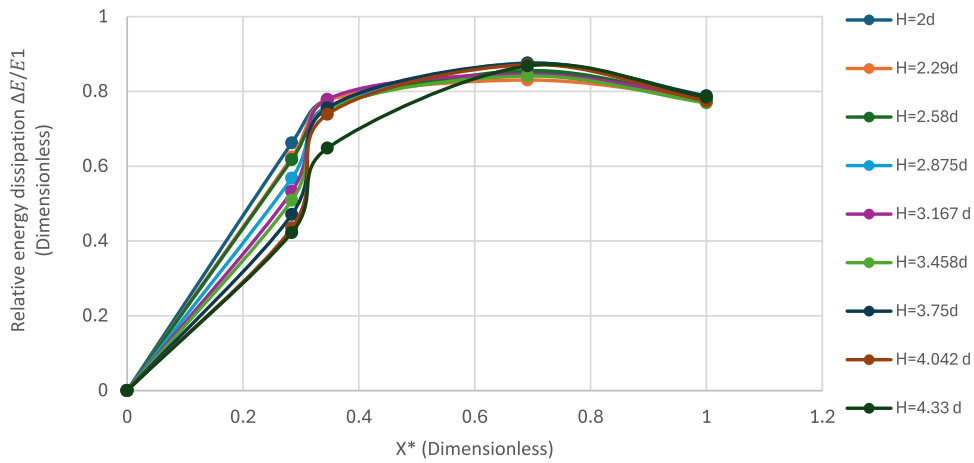


Fig. 10 Relative energy dissipation for different impact wall heights

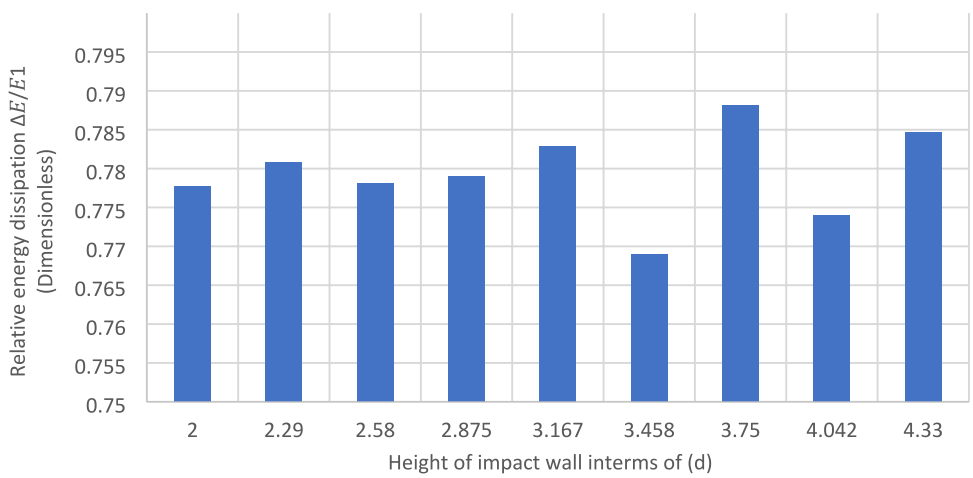


Fig. 11 Relative energy dissipation for different heights of impact wall at end sill of basin for Froude number 3.85

as x* and is a dimensionless parameter that normalizes the position within the basin. For instance, at the end sill, x* takes a higher value, indicating that the data point corresponds to the farthest downstream location in the basin. As depicted in Fig. 6, a higher relative energy dissipation at the end sill of about 0.37% is observed for the in-practice basin compared to the USBR VI basin. The turbulence quantities (turbulent dissipation rate and turbulent

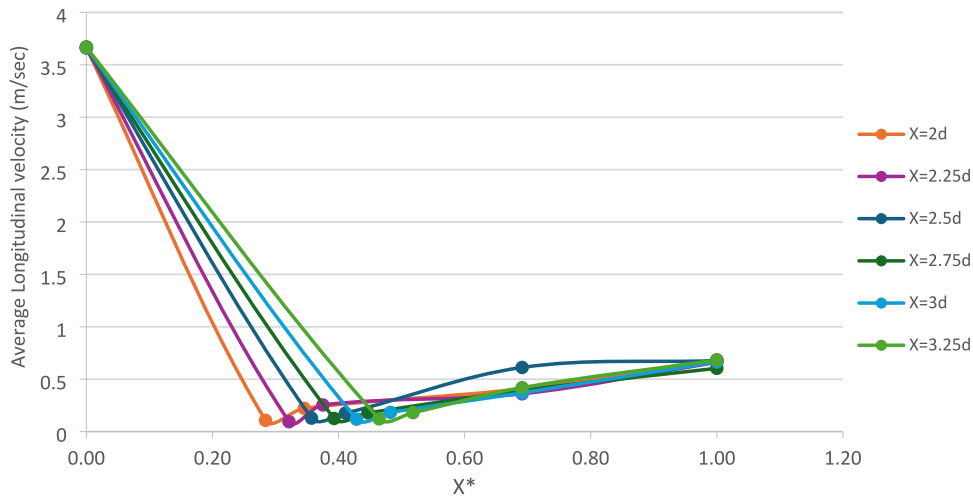


Fig. 12 Average longitudinal velocity changes along different models with various impact wall distances

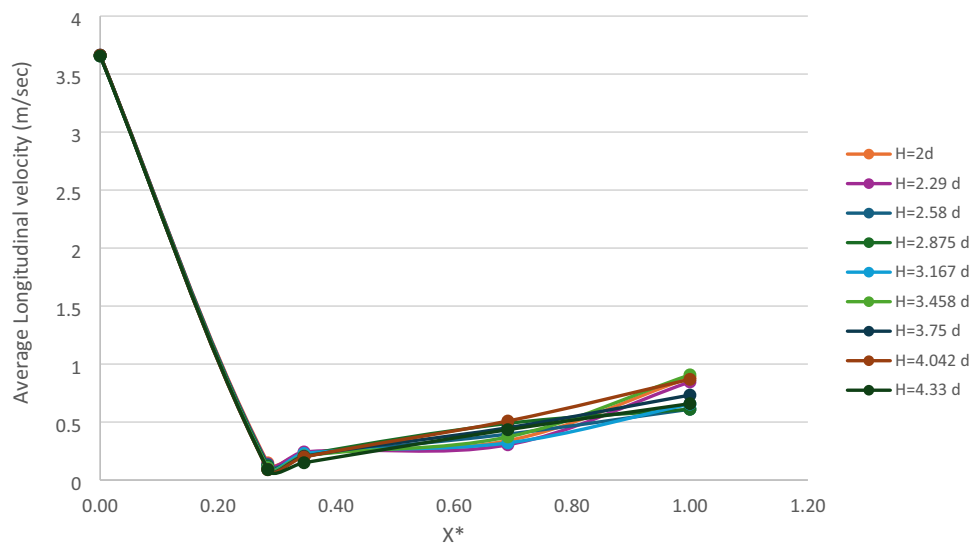


Fig. 13 Average longitudinal velocity changes along different models with various impact wall heights

kinetic energy (TKE)) between the USBR VI basin and the in-practice basin were compared in Fig. 7. Again, the y-axis represents the magnitude of turbulent dissipation rate and turbulent kinetic energy, both of which indicate the intensity of the turbulence generated within the flow. The x-axis still represents the relative position along the basin (x*), showing how turbulence varies from the pipe outlet to the end sill. The resulting turbulent dissipation rate and turbulent kinetic energy shown in Fig. 7 are higher in the later basin.

The results, shown in Fig. 8, indicate that energy dissipation increases as the distance between the impact wall and the pipe outlet decreases, up to a maximum distance of 3d. For the in-practice basin, the highest energy dissipation was recorded at the end sill for impact wall of distance 3d as seen in scenario Z-14, with the relative energy dissipation being approximately 1.5% higher compared to the impact wall at distance 2d as shown in Fig. 9. If the impact wall is placed within or near the pipe entrance (around 3 times the pipe diameter, 3d), the jet impacts the wall with high velocity and momentum, resulting in stronger flow turbulence and mixing. This enhances energy dissipation through turbulent kinetic energy generation and flow breakup. With the wall at the optimal 3d position, the jet impinges directly with minimal spreading, creating strong turbulent eddies and possibly a hydraulic jump near the wall or basin sill.

The water interacts more strongly with the impact wall when it is nearer the pipe outlet, increasing friction and turbulence. Higher energy dissipation is indicated by greater energy loss as a result of this increased interaction. However, the effect of the flow stabilizes after a certain point (3d). Besides, results indicated that variations in the height of the impact wall also influenced energy dissipation. Specifically, higher walls, such as those at 3.167d and 3.75d (scenarios Z-5 and Z-7), led to an increase in energy dissipation, particularly when the impact wall distance exceeded

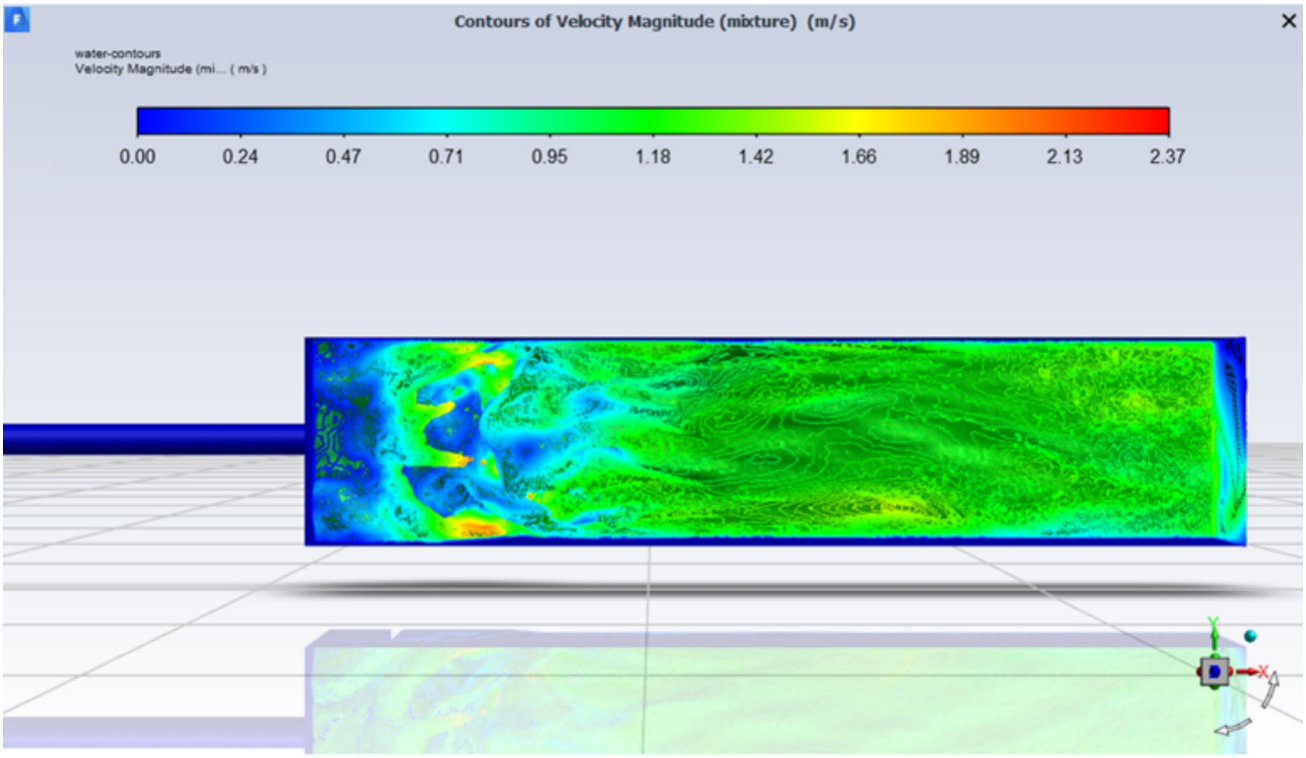


Fig. 14 Velocity contours with streamlines for model Z-5

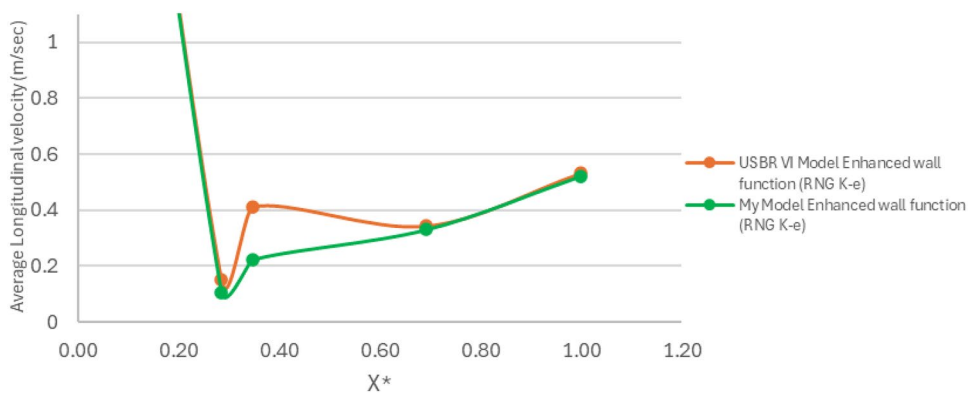
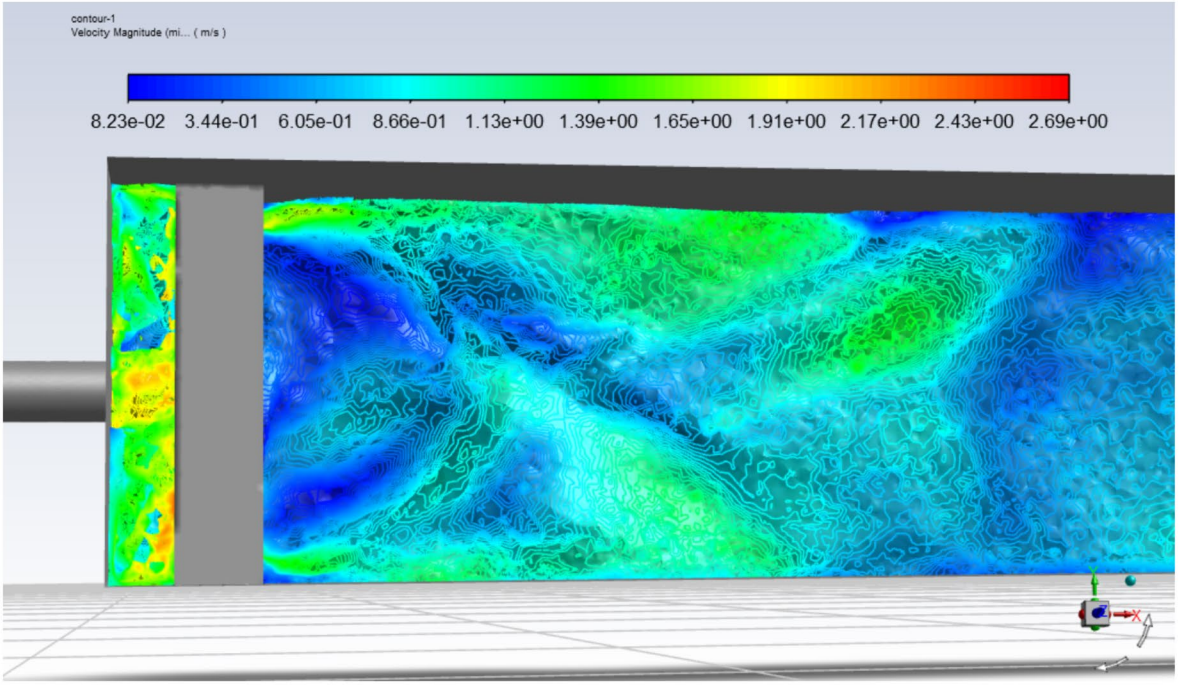
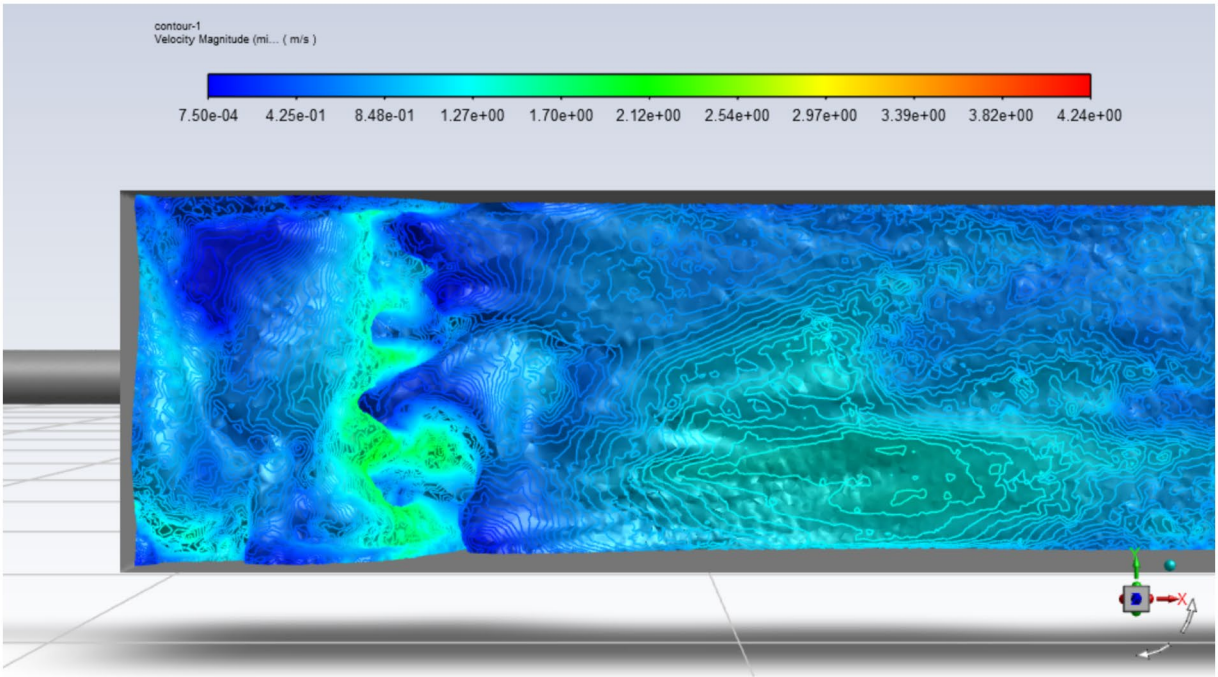


Fig. 15 Average longitudinal velocity changes along USBR VI and Present model

2.5d as shown in Figs. 10 and 11. The flow dissipation is affected by the impact wall's height as it moves through the basin. A greater turbulence and energy loss within the flow is achieved with higher wall, where the vertical interaction of the flow with the wall contributes more to energy dissipation than proximity to the outlet, this is particularly valuable. The maximum amount of energy can be dissipated by a wall placed correctly, not too close and not too far from the outlet, and sufficiently high to ensure maximum dissipation of the flow. A wall that is too short or too far away from the outlet might result in less energy dissipation efficiency. Upon encountering a taller impact wall, the water jet coming from the pipe outlet experiences a notable increase in the vertical extent of its interaction with the wall. This larger vertical surface area imposes a stronger physical barrier to the momentum of the flow, resulting in more pronounced deceleration and redirection of the water. Such obstruction enhances velocity gradients in proximity to the wall, which promote turbulence generation. Furthermore, the increased height of the wall modifies the vertical flow dynamics by compelling the jet to interact with a larger area of the wall's surface, thereby inducing enhanced vertical mixing and flow disruption. This interaction leads to stronger upward and downward velocity fluctuations, encouraging air entrainment



(a)



(b)

Fig. 16 Plan-view contours of velocity magnitude for **a** the USBR VI basin model and **b** the in-practice basin model, highlighting differences in flow velocity distribution and flow behavior between the two designs

and further energy loss. The effect of wall height becomes more pronounced beyond a certain distance from the outlet (greater than 2.5d), where the flow has developed enough to engage fully with the vertical structure.

Longitudinal velocity profies

Following this, the average longitudinal velocity for various configurations of the impact wall positions and heights

are presented in Figs. 12 and 13 The distance between the impact wall and the pipe outlet is the primary variable in Fig. 12, which presents the velocity profiles for different models. The increasing distance of the impact wall is followed by a clear trend in velocity distribution. Higher dissipation occurs in zones near the impact wall due to the sudden disruption in the flow. The velocity reduces further downstream, meaning that the flow has stabilized after sufficiently interacting with the wall. The differences in the impact wall height affect the mean longitudinal velocity, as seen in Fig. 13. The velocity profiles show that a higher impact wall height tends to produce a more turbulent flow, which raises the velocity variation close to the wall.

Water surface velocity contours with streamlines for various arrangements are shown in Fig. 14. It displays the high and low velocity zones and clearly shows the flow pattern downstream and close to the impact wall. The velocity contours also offer a more visual understanding of the velocity distribution, complementing the longitudinal velocity profiles shown in Figs. 12 and 13. The longitudinal velocity profiles of the USBR VI basin and the in-practice basin are indicated in Fig. 15. The velocity profile of the in-practice basin was generally more uniform, with a distinct fall in velocity after the flow interacted with the impact wall.

The plan-view velocity contours for both the USBR VI basin model and the in-practice basin model are depicted in Fig. 16 to illustrate the flow distribution characteristics within the basins. The USBR VI model demonstrates distinct high-velocity zones concentrated near the basin entrance and along the main flow path, reflecting accelerated flow and potential flow separation regions. In contrast, the in-practice basin model displays a more evenly distributed velocity field with reduced peak velocities, indicating a smoother transition and flow stabilization downstream. These variations highlight the impact of design modifications on controlling flow patterns and minimizing localized flow accelerations.

Turbulent dissipation rate and turbulent kinetic energy

Finally, the investigation considers turbulence intensity and related metrics. The longitudinal analysis of turbulent dissipation rate and turbulent kinetic energy (TKE) was carried out to quantify the flow turbulence characteristics for different configurations of impact wall height and distance. Figures 17, 18 and 19 provide insights into the behavior of both turbulent dissipation and TKE in relation to the distance and height of the impact wall. The highest values of both TKE and turbulent dissipation rate were observed near the impact wall, particularly for configuration with distance of 3d, see Figs. 17 and 18. In line with the previously observed increase in energy dissipation, the dissipation rate also peaked at these distances. Longitudinal dissipation rate of turbulence and turbulent kinetic energy (TKE) comparison between varied impact wall heights showed significant variation in both parameters. It was observed that an increase in impact wall height generally resulted in higher turbulent dissipation and TKE.

The peak values of both turbulent dissipation rate and TKE occurred near the impact wall, with the most pronounced increases for wall heights of 3.167d and 3.75d as shown in Fig. 19. Furthermore, while both the longitudinal dissipation rate and TKE decreased downstream, they remained elevated for taller impact walls, suggesting that turbulence persisted over a longer distance.

The plan-view contours of turbulent kinetic energy (TKE) of both the USBR VI basin model and the proposed model are shown in Fig. 20. The outcomes indicate the distribution and magnitude of turbulence throughout the flow field and display prominent variations of the two configurations. The USBR VI model has high TKE values mostly around

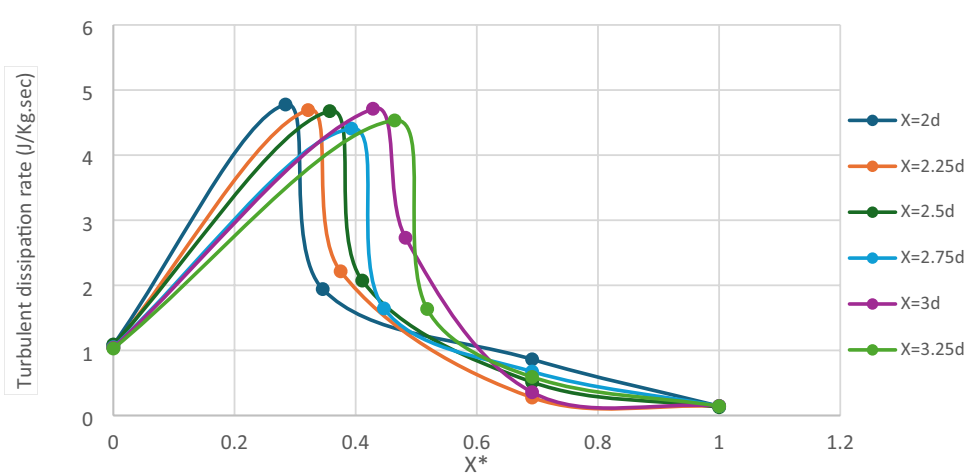


Fig. 17 Longitudinal turbulent dissipation rate for different configurations of distance

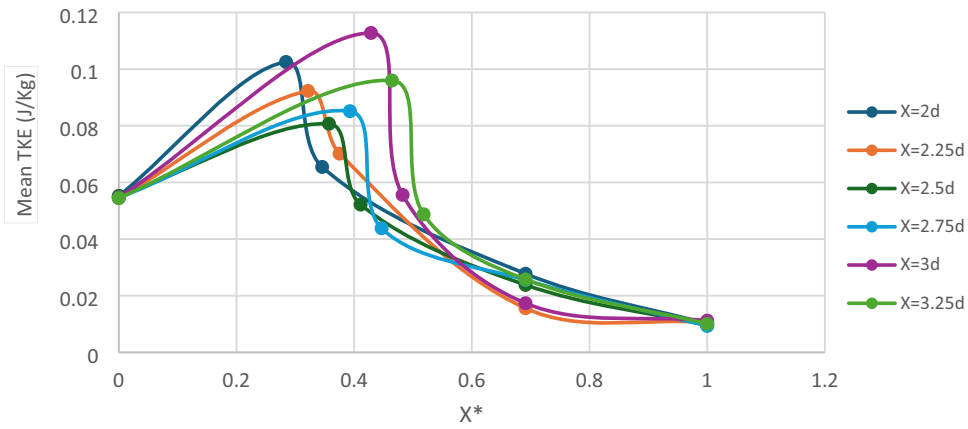


Fig. 18 Longitudinal turbulent kinetic energy for different configurations of distance

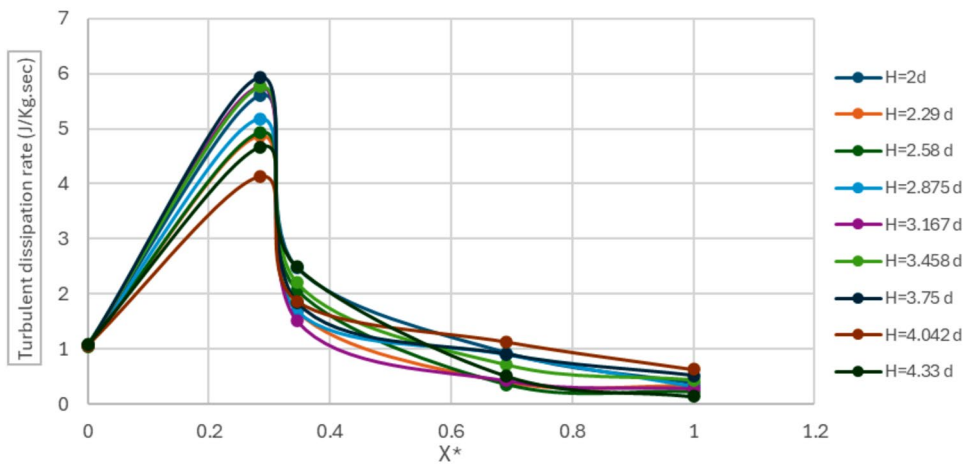


Fig. 19 Longitudinal turbulent dissipation rate for different configurations of height

the basin inlet and the central flow line and reflects intense turbulence resulting from hydraulic jumps and recirculation of the flows. However, the proposed model shows a relative uniform reduction in TKE across critical regions and indicates a higher dissipation of energy as well as better stabilization of the flows.

Limitations of the current study and implications for further studies

The current study only looks at specific impact wall heights and distances influence on energy dissipation. The field applications might have a wider variety of design variables (Wall material or shape) that have an impact on energy dissipation but are not considered in this study. A uniform pipe outlet, smooth flow, and ideal wall geometry are probably assumed in this study. In the field applications, variables such as non-uniform flow rates or surface irregularities in the wall may have an impact on the outcomes. Scaling may have a different effect on energy dissipation than what the model predicts. One notable aspect not explored in the current research is the transport of sediment, which helps determine the energy dynamics of many hydraulic structures. Sediments encounter turbulent eddies, retaining an amount of flow energy which would dissipate hydraulically. The transport and suspension of particles also provide increased resistance to the flow and modify the shape of the channel bed or the riverbed with a subsequent impact on wall roughness and the turbulence structures. Although the study provides valuable insights into optimizing energy dissipation by adjusting impact wall height and distance, further investigation is required to improve the models, test the results in practical imperatives and explore wider implications for various applications.

Conclusion

The findings of this study have significant implications for the design and optimization of pipe outlet stilling basins in various hydraulic applications, particularly in Egypt, such as irrigation systems and hydraulic engineering works, rely on effective energy dissipation. Furthermore, these results are universally applicable and can be adapted to projects

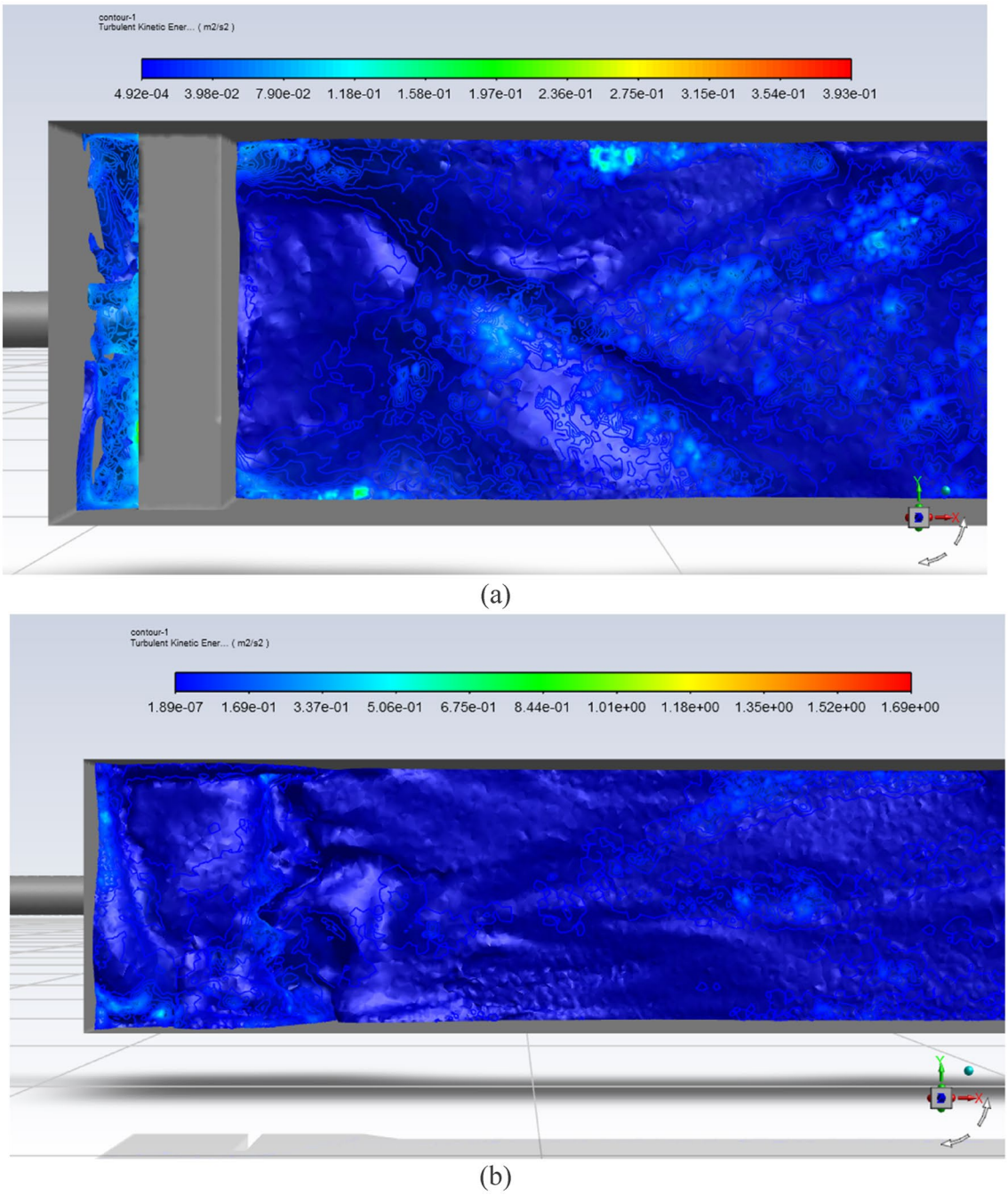


Fig. 20 Plan-view contours of turbulent kinetic energy for **a** the USBR VI basin model and **b** the in-practice basin model, illustrating differences in turbulence intensity and spatial distribution

worldwide, providing a more efficient design for energy dissipation. The USBR VI basin exhibited a more uniform but less pronounced energy dissipation across the various configurations, suggesting that the in-practice basin design, with its adjustable impact wall distances and heights, offers better control over energy dissipation. This is achieved by inducing more localized turbulence and enhancing the conversion of energy into heat. The simulation results reveal that the in-practice basin configuration, especially when the impact wall is positioned closer to the pipe outlet (3d), results in significantly higher energy dissipation rates compared to the USBR VI basin. Furthermore, increasing the height of the impact wall further improved the energy dissipation and reduced the average longitudinal velocity. The turbulent dissipation rate and TKE were most prominent near the impact wall, with a clear dependence on both the distance

and height of the impact wall. Overall, the in-practice basin design proves to be more efficient in dissipating energy, thereby enhancing the hydraulic performance of pipe outlet stilling basins. Therefore, the results can be summarised as follows: -

- The in-practice basin proves to be efficient in dissipating energy when compared to conventional USBR VI basin with a higher relative energy dissipation at end sill of about 0.37%.
- The best position of impact wall based on current simulations is at distance 3d from pipe outlet where relative energy dissipation for impact wall at distance 3d improves by an average of 1.5% compared to 2d.
- Higher impact walls (3.167d and 3.75d) dissipate more energy particularly when the impact wall distance exceeded 2.5d
- The longitudinal velocity profile, turbulent dissipation rate and turbulent kinetic energy strengthen the highlighted results.
- The peak values of both turbulent dissipation rate and TKE occurred near the impact wall, with the most pronounced increases for wall heights of 3.167d and 3.75d.

Declarations

Conflict of interest The authors declare there is no conflict of interest.

References

Abbas, A., Alwash, H., & Mahmood, A. (2018). Effect of baffle block configurations on characteristics of hydraulic jump in adverse stilling basins. *MATEC Web of Conferences*. https://doi.org/10.1051/matecconf/201816203005

Al-Husseini, T. R., Al-Yousify, H. A., & Al-Ramahee, M. A. (2018). Experimental and numerical study of the effect of the downstream spillway face's angleon the stilling basin's energy dissipation. *International Jounal of Civil Engineering Technology, 9*(8), 1327–1337.

Alghwail, A., Abdusalam, D., Stević, S., & Abourohiem, M. A. J. G. (2018) Dissipation of mechanical energy over spillway through counter flow, *70*(05), 377–391.

ANSYS 2016. ANSYS Fluent—CFD Software. ANSYS.

ANSYS, i. (2009). General documentation theory guide 12.0.

Arjenaki, M. O., & Sanayei, H. R. Z. (2020). Numerical investigation of energy dissipation rate in stepped spillways with lateral slopes using experimental model development approach. *Modeling Earth Systems and Environment, 6*(2), 605–616.

Arun Goel, A., & Verma. (1999). Improved design of energy dissipators for pipe outlets, *13*(4), 313–320.

Atkins, P. W. (1994) *The Second Law.* Scientific American Library.

Babaali, H., Mojtahedi, A., Soori, N., & Soori, S. (2019). Numerical modeling of flow in USBR II stilling basin with end adverse slope. *International Journal of Environmental Ecological Engineering, 13*(2), 62–68.

Behnamtalab, E., Ghodsian, M., Zarrati, A., & Salehi Neyshabouri, S. A. A. (2019). Geometry modification of stilling basin USBR VI with numerical simulation. *Journal of Hydraulics, 13*(4), 1–15.

Behnamtalab, E., Lakzian, E., & Hosseini, S. (2022). *Study of USBR VI stilling basin with entropy generation index* (pp. 1–16).

Behnamtalab, E., Maskani, V., & Sarkardeh, H. (2023). Numerical study of turbulent flow in USBR VI stilling basin. *Applied Water Science, 13*(7), 146.

Beichley, G. (1971). Hydraulic design of stilling basin for pipe or channel outlets.

Brevard, J. A. (1971). Criteria for the hydraulic design of impact basins associated with full flow in pipe conduits.

Chanson, H. (2000). Historical development of stepped cascades for the dissipation of hydraulic energy. *Transactions of the Newcomen Society, 72*(2), 295–318.

Chanson, H. (2002). *The hydraulics of stepped chutes and spillways*. Balkema Publishers.

Choi, D., Sabnis, J., & Barber, T. Applications of an RNG k-epsilon model to compressible turbulent shear layers. *32nd Aerospace Sciences Meeting and Exhibit*, *188*.

Corry, M. (1975). *Hydraulic design of energy dissipators for culverts and channels.* US Department of Transportation, Federal Highway Administration.

Experimental Validation of CFD Modeling of Multiphase Flow through Open Channel Confluence. *World Environmental and Water Resources Congress 2011* (pp. 2176–2183).

Farouk, M., Kriaa, K., & Elgamal, M. (2022). CFD simulation of a submersible passive rotor at a pipe outlet under time-varying water jet flux. *Water, 14*(18), Article 2822.

Fiala, G. R., & Albertson, M. L. (1963). Manifold stilling basin. *Transactions of the American Society of Civil Engineers, 128*(1), 428–454.

Flammer, G. H., Skogerboe, G. V., Wei, C.-Y., & Rasheed, H. (1970). Closed conduit to open channel stilling basin. *Journal of the Irrigation Drainage Division, 96*(1), 1–10.

Garde, R., Ranga Raju, K., & Sujudi, A. (1990). Design of settling basins. *Journal of Hydraulic Research, 28*(1), 81–91.

Gehlot, B. K., & Tiwari, H. (2014). Critical review of stilling basin models for pipe outlet works. *International Journal of Research in Engineering Technology*, 2319–1163.

Goel, A. (2008). Design of stilling basin for circular pipe outlets. *Canadian Journal of Civil Engineering, 35*(12), 1365–1374.

Goel, A., & Verma, D. (2006). Alternate designs of stilling basins for pipe outlets. *Irrigation Drainage Systems, 20*(1), 139–150.

Hager, W. (2018). *Trajectory basins* (pp. 111–150). Routledge.

Hager, W. H. (1994). Countercurrent jet device. *Journal of Hydraulic Engineering, 120*(4), 504–517.

Hager, W. H. (2010). Hydraulic jump and stilling basins. In W. H. Hager (Ed.), *Wastewater hydraulics: Theory and practice* (pp. 174–215). Springer Berlin Heidelberg.

Hager, W. H. (2013). *Energy dissipators and hydraulic jump.* Springer Science & Business Media.

Hager, W. H., & Dams. (1996). Performance of a counter-current type stilling basin. *The International Journal on Hydropower, 3*(4), 78–84.

Jain, S., & Jacob, A. (1970). Centre-state relations in water resources development. *Journal of the Indian Law Institute, 12*(1), 1–25.

Jeong, W., & Seong, J. (2014). Comparison of effects on technical variances of computational fluid dynamics (CFD) software based on finite element and finite volume methods. *International Journal of Mechanical Sciences, 78*, 19–26.

Kamaraj, M., & Rangarajan, S. J. (2022). Predicting the future land use and land cover changes for Bhavani basin, Tamil Nadu, India, using QGIS MOLUSCE plugin. *Environmental Science Pollution Research, 29*(57), 86337–86348.

Kannan, B., Karthikeyan, S., & Sundararaj, S. (2016). Comparison of turbulence models in simulating axisymmetric jet flow. In *Innovative design and development practices in aerospace and automotive engineering: I-DAD, February 22–24, 2016* (pp. 401–407). Springer.

Kannan, B. T., Karthikeyan, S., & Senthilkumar, S. (2017). Comparison of turbulence models in simulating axisymmetric jet flow. In *Innovative design and development practices in aerospace and automotive engineering* (pp. 401–407). Springer Singapore.

Keim, S. R. (1962). The contra costa energy dissipator. *Journal of the Hydraulics Division, 88*(2), 109–122.

Kheirehdast, S., & Ghodsian, M. (2024). Effect of endsill shape on scouring depth and index downstream of USBR type VI stilling basin. *Modares Civil Engineering Journal, 24*(2), 19–29.

Lu, L., Li, Y.-C., Teng, B., & Chen, B. (2008). Numerical simulation of turbulent free surface flow over obstruction. *Journal of Hydrodynamics, 20*(4), 414–423.

Mardani, M., Rahimzadeh, H., & Sarkardeh, H. (2015). Analysis and assessment of installing blocks on performance of stilling basins. *Modares Mechanical Engineering, 15*(6), 31–41.

Mei, J., Zhou, Y., Xu, K., Xu, G., Shu, Z., Gan, Q., Yu, Q., & Qian, J. (2025). Energy dissipation on inclined stepped spillways. *Water, 17*(2), Article 251.

Mohammed, S. R., Nile, B. K., & Hassan, W. H. (2020). Modelling stilling basins for sewage networks. *IOP Conference Series: Materials Science and Engineering*. https://doi.org/10.1088/1757-899X/671/1/012111

Pagliara, S., Roshni, T., & Palermo, M. (2015). Energy dissipation over large-scale roughness for both transition and uniform flow conditions. *International Journal of Civil Engineering, 13*(3), 341–346.

Papageorgakis, G., & Assanis, D. N. (1999). Comparison of linear and nonlinear RNG-based k-epsilon models for incompressible turbulent flows. *Numerical Heat Transfer, Part b: Fundamentals, 35*(1), 1–22.

Paul Peter Urone, R. H. (2022). College Physics 2e.

Pillai, N. N., Goel, A., & Dubey Ashoke, K. (1989). Hydraulic jump type stilling basin for low Froude numbers. *Journal of Hydraulic Engineering, 115*(7), 989–994.

Rasheed, H. (1963). Utah State University stilling basin pipe flow to open channels.

Rehman, K., & Hong, S. H. (2022). Influence of lateral flow contraction on bed shear stress estimation by using measured turbulent kinetic energy. *Experimental Thermal and Fluid Science, 139*, Article 110742.

Riad, P. (2008) *A study of the hydraulic performance of stilling basins using physical and numerical models (Delft 3D), case study: New Naga Hammadi Barrage on The Nile.*

Rice, C. E., & Kadavy, K. C. (1993). Protection against scour at SAF stilling basins. *Journal of Hydraulic Engineering, 119*(1), 133–139.

Salmasi, F., & Abraham, J. (2022a). Effect of slope on energy dissipation for flow over a stepped spillway. *Water Supply, 22*(5), 5056–5069.

Salmasi, F., & Abraham, J. (2022b). Genetic algorithms for optimizing stepped spillways to maximize energy dissipation. *Water Supply, 22*(2), 1255–1274.

Shahheydari, H., Nodoshan, E. J., Barati, R., & Moghadam, M. A. (2015). Discharge coefficient and energy dissipation over stepped spillway under skimming flow regime. *KSCE Journal of Civil Engineering, 19*, 1174–1182.

Sharma, I., Mishra, A., & Mehrotra, R. (2021). Performance evaluation of impact stilling basin using ANSYS fluent. In *Advances in water resources and transportation engineering* (pp. 139–149). Springer.

Sharma, S., Jesudhas, V., Balachandar, R., & Barron, R. (2019). Turbulence structure of a counter-flowing wall jet. *Physics of Fluids*. https://doi.org/10.1063/1.5082550

Sobhan Aleyasin, S., Fathi, N., & Vorobieff, P. (2015). Experimental study of the type VI stilling basin performance. *Journal of Fluids Engineering*. https://doi.org/10.1115/1.4029164

Stevović, S. A., & Amad (2016). Management and mitigation of erosion process effects on the protection length of stilling basins by means of a reversed jet flow.

Tiwari, H. (2013). Design of stilling basin model with impact wall and end sill. *Research Journal of Recent Sciences, 2277*, 2502.

Tiwari, H., & Goel, A. (2014). Effect of end sill in the performance of stilling basin models. *American Journal of Civil Engineering Architecture, 2*(2), 60–63.

Tiwari, H., & Goel, A. (2016). Effect of impact wall on energy dissipation in stilling basin. *KSCE Journal of Civil Engineering, 20*(1), 463–467.

Tiwari, H., Goel, A., & Gahlot, V. (2011). Experimental study of effect of end sill on stilling basin performance. *International Journal of Engineering Science Technology, 3*(4), 3134–3140.

Tiwari, H., Goel, A., Sharma, A., & Balvanshi, A. (2022). *Performance improvement of Usbr VI stilling basin model for pipe outlet* (pp. 1–8). Springer.

Tiwari, H., Goel, A., Suresh, S., & Tiwari, S. (2015). Effect of inverted t-shape splitter blocks on the performance of stilling basin models. *Aquatic Procedia, 4*, 1561–1568.

Tiwari, H., Pawar, A., Gehlot, B., & Singh, J. (2014). Study of shape of intermediate sill on the design of stilling basin model. *International Journal of Research in Egg Technology, 3*(4), 133–138.

Tiwari, H., & Seema, T. (2013). Design of stilling basin models with intermediate sill. *Journal of Science & Technology Management, 2*(4), 66–71.

Tiwari, H., & Seema, T. Reduction of scour depth downstream of pipe outlet stilling basin using end sill.

Tiwari, H. L., & Singh, K. (2017). Experimental study of stilling basin models.

Varshney, R. J. W., & International, E. (1965). Stilling basin characteristics for low froude numbers, *22*(2), 157–172.

Verma, D., & Goel, A. (2003). Development of efficient stilling basins for pipe outlets. *Journal of Irrigation and Drainage Engineering, 129*(3), 194–200.

Vischer, D. (2018). *Types of energy dissipators* (pp. 9–21). Routledge.

Vollmer, E., & Khader, M. (1971). Counter current energy dissipator for conduit outlets. *International J. of Water Power, 23*(7), 260–263.

Wang, L., Li, Z., & Diao, M.-J. (2022). Hydraulic characteristics of countercurrent jets on adverse-sloped beds. *Water Supply, 22*(2), 1876–1888.

Wei-Lin, X., Hua-Sheng, L., Yong-Quan, Y., & Chi-Gong, W. (1996). Numerical simulation of 3-D turbulent flows of plunge pool and energy dissipation analysis. *Journal of Hydrodynamics, 5*.

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